batteries connected in series can be different and batteries may need to be charged with different currents, up to different maximum voltages and for different time durations. Unlike many chargers, this charger can be used to charge any number of batteries from one to four simultaneously. The charging current for all the batteries can be monitored individually with voltmeters connected in parallel across resistors R1 through R4. Connectors CON2 through CON5 serve this purpose.

The approximate charging current can be calculated as:

I = (measured voltage/30 ohms) amperes

Here, suppose you measure voltage U1 = 2.7V over resistor R1, U2 = 3V over resistor R2, U3 = 2.4V over resistor R3 and U4 = 3V over resistor R4. With these values, you get charging currents as I1 = 90mA, I2 = 100mA, I3 = 80mA and I4 = 100mA. The charging current depends on the electric charge available in the battery. Refer the respective datasheets of batteries in order to know their charging current and the maximum allowed voltage.

The batteries can be of different types and with different initial charge because each battery is charged individually. The status of all the batteries is monitored by individual comparators in IC2 and the results displayed individually by LED5 through LED8.

Using preset VR1, you can set the reference voltage of all the four comparators to an appropriate level, say, 1.5V. The maximum reference voltage (2.5V) is set by IC1. Comparator A4 of IC2 monitors Batt.1, A3 monitors Batt.2, A2 monitors Batt.3 and A1 monitors Batt.4. So, for example, if the voltage of Batt.1 is higher than the preset voltage, LED8 will glow.

Discharger. Sometimes you need to manually discharge the rechargeable batteries. That can be done easily with this circuit. First, remove the power supply from the circuit. Next, close the respective jumpers. For example, if you wish to discharge Batt.1 and Batt.2, close only J1 and J2. The batteries start discharging through electric bulbs B1 and B2. You may connect voltmeters across CON6 and CON7 to check the voltage status on the corresponding discharging batteries.

To drain the batteries, choose bulbs with appropriate ratings. In most cases, bulbs rated 2.5V/200mA or 2.5V/300mA will do the job. You may also use 3.5V/300mA or 6.3V/300mA bulbs but these produce dim light. Open the jumpers J1 through J4 when you charge the batteries.

This battery charger does not need any special adjustment to operate properly. You only need to set the comparator threshold at 1.5V or any other appropriate power supply. Equalisation

PARTS LIST

Semiconductors:	
IC1	- TL431 shunt regulator
IC2	- LM324 quad op-amp
D1-D4	- 1N4007 rectifier diode
LED1-LED9	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):	
R1-R4	- 30-ohm, 1-watt
R5-R8	- 300-ohm
R9-R13	- 10-kilo-ohm
R14-R17	- 470-kilo-ohm
R18-R21	- 2.2-mega-ohm
R22-R25	- 3.3-kilo-ohm
R26	- 2.2-kilo-ohm
R27-R30	- 33-ohm
VR1	- 10-kilo-ohm preset
Capacitors:	
C1	- 100nF ceramic disk
C2	- 100µF, 25V electrolytic
Miscellaneous:	
J1-J4	- Shorting jumper with 2-pin male bergstrip
Batt.1-Batt.4	- 1.2V rechargeable battery
B1-B4	- 2.5V, 200mA bulb
CON1	- 2-pin terminal connector
CON2-CON9	- 2-pin connector
CON10-CON11	- USB A-type connector with cable
	- 5V DC regulated/USB power supply